

Cover letter for submission to Environmental Research Letters

Title: Power spectrum scaling as a measure of critical slowing down and precursor to tipping points in dynamical systems.

Authors: Joshua Prettyman, Tobias Kuna (University of L'Aquila, Italy; Mathematics of Planet Earth CDT, University of Reading, UK), Valerie Livina (National Physical Laboratory, London).

We are pleased to submit the enclosed manuscript for consideration. The author Valerie Livina has been invited by Lauren Carter to submit to the upcoming focus issue "Focus on Earth System Resilience and Tipping Behaviour" and the manuscript has been prepared with this in mind. A waiver of the usual publication fee has been agreed with Lauren Carter in advance due to the current situation in which the authors are unable to access funding via an academic institution.

The content of this paper builds on work completed for the PhD thesis of Joshua Prettyman and has here been prepared for a wider audience outside the narrative of the thesis manuscript.

The work studies the power spectrum and discrete periodogram of the AR(1) process which has previously been used as a model of the critical slowing down phenomenon. By doing so we are able to justify the use of the power-spectrum (PS) scaling exponent as a tipping point indicator. We also extend this approach to study several properties of the PS indicator including its robust response to the presence of trends and periodicities in the system. Both of these aspects to the paper are, to our knowledge, novel research questions and yield useful contributions to the field of tipping points and resilience in dynamical systems.